Задача 4. (Pippard 172) Three identical bodies of constant thermal capacity are at temperatures 300 K, 300 K, and 100 K. If no work or heat is supplied from outside, what is the highest temperature to which any one of these bodies can be raised by the operation of heat engines?
Задача 5. Земята е плоска и нейната атмосфера е изотермна.
Дадено: g – земно ускорение, m – маса на частица въздух, k – константа на Болцман, T – температура на атмосферата, n_0 – концентрация на атмосферата на повърхността на Земята.
а) Търси се: $n(h)$ – броя частици на единица обем като функция на височината на атмосферата.
Частиците от газа с тази температура се движат с най-различни скорости, които обаче се подчиняват на някакво конкретно разпределение – съществува функция $f(v_x)$, така че броят частици в обема dV, чиято скорост е в интервала $(v_x,\ v_x{+}dvx)$, са $dN = nf(v_x)dv_x dV$.
б) Приемайки, че частиците се движат свободно и единствено се удрят еластично в повърхността на Земята, намерете $f(v_x)$.
Разпределението на молекулите по останалите компоненти на скоростта е същото.
в) Намерете разпределението на молекулите по големина на скоростта.
г) Намерете средноаритметичната големина на скоростта, най-вероятната големина на скоростта и средноквадратичната скорост.
Задача 6. (IPhO 2018.3B) Tumour growth (5.5 points)

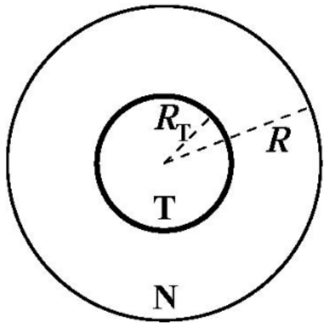

Figure 3. Simplified tumour.

Tumour growth is a very complex process where biological mechanisms such as cell proliferation and natural selection are intertwined with physics. In this problem we will consider a simplified model of tumour growth that addresses the increase in pressure commonly observed in solid tumors.
Consider a group of normal cells forming a tissue surrounded by an inextensible basement membrane, which forces the tissue to maintain always the same form: a sphere of radius R (Figure 3).
Initially the tissue does not have residual stresses, i.e. the pressure at every point is equal to the atmospheric pressure.
At time $t = 0$, a tumour starts growing at the centre of this sphere and, as it grows, the pressure inside the tissue increases. Consider that both tissues (normal, N, and tumour, T) are compressible such that their densities, ρN and ρT, increase linearly with pressure:

$$\varrho_N = \varrho_0\left(1+\frac{p}{K_N}\right), \qquad \varrho_T = \varrho_0\left(1+\frac{p}{K_T}\right),$$

where ρ_0 is the rest tissue density, p is the pressure difference to the atmospheric pressure and K_N , K_T are the compressibility moduli (bulk moduli) of the normal and tumour tissues, respectively. In general, tumours are stiffer and so they have a higher bulk modulus.
B.1 The mass of normal cells is not altered while the tumour is growing. Obtain the ratio between the tumour volume and the total tissue volume, $v = V_T/V$, as a function of the ratio between the tumour mass (M_T) and the normal tissue mass (M_N), $\mu = M_T/M_N$ and the ratio of the bulk moduli, $\varkappa = K_N/K_T$. [1.0 pt]
Hyperthermia is sometimes used together with chemotherapy and radiotherapy in the treatment of cancer. In hyperthermia the cancer cells are selectively heated from the normal human body temperature, 37 °C, to temperatures above 43 °C, inducing their death. Researchers are currently developing carbon nanotubes covered with special proteins capable of binding to tumour cells. When the tissue is irradiated with near-infrared radiation, the nanotubes absorb it in a much greater extent than the surrounding tissues and therefore can be selectively heated as well as the tumour cells to which they are attached.

Consider that the tumour, the normal cells and the surrounding tissue have a constant thermal conductivity k, i.e. in the geometry of this problem, the energy that crosses a spherical surface of radius r per unit time and per unit area is equal to k times the derivative of the temperature with respect to r. The nanotubes are uniformly distributed in the tumour volume and are able to deliver a power $\mathscr{P}$ of thermal energy per unit volume. Assume that the temperature is equal to the normal human body temperature very far away from the tumour.

B.2 Obtain, for the stationary state, the temperature at the centre of the tumour as a function of $\mathscr{P}$, k, the human body temperature and the tumour radius, R_T. [1.7 pt]

B.3 Obtain the minimum power per unit volume, $\mathscr{P}_{\min}$, needed to heat up all tumour cells in a tumour with 5.0 cm radius to a temperature larger than 43.0 °C. Take the thermal conductivity of the tissue to be equal to $k = 0.60$ W K^{-1}m^{-1}. [0.5 pt]

Consider that the tumour is irrigated by a vessel network with a branched structure like in question A.1. As the tumour grows, when its pressure p becomes larger than the pressure P_{cap} at the thinnest vessels, the radii of these vessels will decrease by a small amount δr. If this pressure reaches a critical value p_c (which would correspond to a radius decrease of δr_c), the thinnest vessels would collapse, compromising seriously the irrigation to the tumour. The pressure and the radius change can be related by the following phenomenological relation:

$$\frac{p}{P_{\text{cap}}}-1=\left(\frac{p_c}{P_{\text{cap}}}-1\right)\left(2-\frac{\delta r}{\delta r_c}\right)\frac{\delta r}{\delta r_c}.$$

Consider that just the smallest vessels (of level $N-1$) have their radius altered when the tumour increases its pressure.

B.4 In the linear regime (i.e. consider that $p - P_{\text{cap}}$ is very small), express the relative drop in the flow rate, $\dfrac{\delta Q_{N-1}}{Q_{N-1}}$, in these thinnest vessels, as a function of the tumour volume ratio $v = V_T/V$ and K_N, N, p_c, δr_c, r_{N-1}, P_{cap}. [2.3 pt]